

Manuscript title:

Restrictions on fronting and stress shift in the encoding of Czech focus

Submitted for review at *Journal of Slavic Linguistics*

Abstract: This article examines two operations which are used to encode focus in Czech: word order alternation and stress shift. Based on data from a previous production experiment, we ran a rating study where participants rated the authenticity of utterances with manipulated information structure. The results showed that alternating word order to encode focus is free, supporting the claims about the flexibility of Czech word order. Shifting stress is a more costly operation – it is possible (in terms of competence), but it is not necessarily easy (in terms of performance). Moreover, applying a word order alternation and stress shift at the same time (in object fronting cases) leads to a significant decrease in rating, which cannot be accounted for by a mere cumulative effect of noncanonical word order and noncanonical stress. We argue instead that the noncanonical structure is blocked by a more economical one, where the same information structure is conveyed with the default structure and prosodic pattern.

keywords: focus · Czech · word order · stress shift · information structure

This file was last updated on July 20, 2026

1 Introduction

Cross-linguistically, information structural concepts such as focus or givenness have been observed to be encoded by several means: word order alternation, prosodic marking, morphology, particles, or syntactic constructions such as clefts. In this article we zoom in on the first two encoding strategies in Czech, a language which has been claimed to exhibit both flexible word order and flexible prosody. This is illustrated by example (1), from Groeben et al. (2017, 186), who supplied evidence from an acceptability rating experiment that subject focus may be encoded equally well by stress shift (from the default final position to the marked initial position), see (1-B₁), and by changing the word order (from the default SVO to the marked OVS), see (1-B₂).¹

- (1) A: ‘Who drank alcohol on that day?’
- B₁: [Josef NĚMEC]_F pil ten den alkohol.
Josef NĚmec drank DEM day alcohol
- B₂: Alkohol pil ten den [Josef NĚMEC]_F.
alcohol drank DEM day Josef NĚmec
- For both: ‘Josef NĚmec drank alcohol on that day.’

The more nuanced experiments of Šimík & Wierzbica (2017) have shown that both stress shift and word order alternations (aka scrambling) come with a certain cost (reflected in an acceptability decrease), whereby the former appears to be a bit more costly than the latter. This can be illustrated by example (2) (from Šimík & Wierzbica 2017, 688), where the need to destress the discourse given object is achieved more naturally by using the noncanonical SOV order, see (2-B₁), than by shifting the stress to the verb, see (2-B₂). Still, both are more acceptable than any condition with the object stressed, showing the strength of the requirement to destress given elements.

- (2) A: ‘We’d like to talk to the Italian ambassador.’
- B₁: Myslím, že pan Sládek [toho velvyslance]_G KONTAKTOVAL.
think that Mr Sládek DEM ambassador contacted
- B₂: Myslím, že pan Sládek KONTAKTOVAL [toho velvyslance]_G.
think that Mr Sládek contacted DEM ambassador
- For both: ‘I think that Mr. Sládek has contacted the ambassador.’

Šimík & Wierzbica (2017) argue that the differences in acceptability do not reflect grammaticality distinctions, but rather preferences. They also assume that constraint violations, e.g. the deviation from the canonical stress or word order pattern, contribute cumulatively to

¹The word that is set in SMALL CAPS is the locus of the sentence stress. The subscripts F and G indicate focus and givenness, respectively.

the overall acceptability decrease. That is, if stress shift causes a penalty x and scrambling a penalty y , the overall penalty assigned to a clause where both operations have applied would be $x + y$. The authors notice that this cumulativity assumption does not always seem to be sufficient. More specifically, the acceptability decrease caused by shifting the stress to a scrambled focused object, as in (3) (from Šimík & Wierzba 2017, 688; our judgement added), is considerably more dramatic than what one would expect by a simple cumulation of the penalties caused by shifting the stress and by scrambling.

- (3) A: ‘Do you have an idea who Mr. Sládek contacted in the end?’
 B: ?Myslím, že pan Sládek [toho VELVYSLANCE]_F kontaktoval.
 think that Mr Sládek DEM ambassador contacted
 Intended: ‘I think that Mr. Sládek has contacted the ambassador.’

Šimík & Wierzba (2017) hypothesize that the unacceptability of (3) could reflect a specific constraint on the scrambling of focused elements to a non-peripheral position, known from literature on German (aka A-scrambling); see Lenerz (1977). We note that, intuitively, the scrambling of a focused object to the clausal periphery (aka A-bar scrambling) is more acceptable:

- (4) A: ‘Do you have an idea who Mr. Sládek contacted in the end?’
 B: Myslím, že [toho VELVYSLANCE]_F pan Sládek kontaktoval.
 think that DEM ambassador Mr Sládek contacted
 ‘I think that Mr. Sládek has contacted the ambassador.’

In this paper, we compare the acceptability of stress shift to focused subjects, as in (1), with the acceptability of stress shift to fronted (A-bar-scrambled) focused objects, as in (4). Assuming that there is a qualitative difference between A- and A-bar scrambling, we consider two hypotheses: (i) Stress shift to a fronted object will exhibit a cumulative penalty of stress shift and fronting (following the spirit of Šimík & Wierzba’s 2017 proposal). (ii) The penalty associated with a stress shift to a fronted object will be bigger than what one would expect from a cumulative effect of two constraint violations; the rationale behind this hypothesis (following the spirit of Reinhart 2006) is that fronting a focused object involves two costly operations in a situation where neither is in principle required: the same effect can be achieved by using a canonical stress pattern (final stress) and canonical word order (SVO). Our experimental results support hypothesis (ii).

In addition, we will show that while stress shift is generally available in Czech grammar, it is not easy to produce for less proficient native speakers in an experimental setting. We interpret this as an indication of the relative strength of the requirement to place sentence stress clause-finally (aka nuclear stress rule; see below), already observed in Šimík & Wierzba (2017).

1 The article is organized as follows. In section 2 we summarize previous claims about
 2 information structure in Czech and its encoding in word order and prosody as well as
 3 introduce the relevant constraints that govern linguistic form, especially word order and
 4 sentence prosody. Section 3 presents the rating experiment which was designed to compare
 5 the effects of the constraints on the perceived naturalness of utterances. We describe the
 6 procedure, design and materials as well as the statistics used to analyze the results. We
 7 discuss the findings in section 4.

8 2 Background

9 Czech word order is often said to be free (Siewierska & Uhlřová, 1998). As opposed to
 10 English for example, utterances may exhibit various orderings of the subject, verb and
 11 object. However, it has often been claimed that the order of Czech words is conditioned by
 12 certain categories of information structure (= IS).² For example, Kučerová (2011) proposes
 13 that in Czech given constituents must precede new ones and analyzes this phenomenon as
 14 an effect of a specific operator called G. A number of observations has been made about
 15 the syntactic marking of focus. Traditionally, it has been claimed that focus in Czech
 16 tends to be placed utterance-finally (Mathesius, 1939; Sgall et al., 1980; Firbas, 1992).
 17 The role of word order has been predominant in the discussions of IS encoding in Czech so
 18 far. It should be noted that the previous accounts use different definitions of focus (calling
 19 it also rheme or rheme proper) and givenness (either based on the notion of importance,
 20 highlighting or contextual boundness), which differs from our approach described below.

21 Similarly to word order, prosody in Czech is claimed to be flexible. The role of prosody
 22 in Czech IS encoding has been emphasized by e.g. Mathesius (1939) or Daneš (1957). As
 23 for focus, they both notice that although a focused constituent can occur in any position in
 24 the utterance, it requires prosodic prominence. In this way Czech focus behaves similarly
 25 to other Slavic languages (for Russian: Alter 1997a,b; Zybatow & Mehlhorn 2000; for
 26 Polish: Hamlaoui et al. 2018; for a recent survey, see Šimík 2024).

27 In the present article, we consider IS notions as binary oppositions, based on the
 28 account in Krifka (2008). Namely, we work with the oppositions of focus-background
 29 and given-new. A focused constituent gives rise to alternative denotations relevant in the
 30 current discourse (Rooth, 1992); background is what is not focused. A given constituent
 31 has a synonymic or hyponymic antecedent in the immediately preceding discourse; new
 32 material is what is not given.

33 That focus bears prosodic prominence has been observed before (Jackendoff, 1972;

²Apart from the IS factors, other regularities have been observed, e.g. the fixed placing of clitics in the second position (Wackernagel, 1892).

¹ Truckenbrodt, 1995; Zubizarreta, 1998; Reinhart, 2006).³ This is usually captured by
² the Stress Focus (SF) constraint, provided in (5). Recently there has been a number of
³ empirical studies where the prosodic nature of focus was tested (Baumann et al. 2006;
⁴ Breen et al. 2010; Kügler & Gollrad 2015, a.o.; for Czech: Šimík & Wierzba 2017; Hamlaoui
⁵ et al. 2018). Givenness, on the other hand, leads to (relative) deaccentuation, as captured
⁶ by the constraint in (6). This tendency was also tested empirically, e.g. by Baumann &
⁷ Riester (2013). It is important to note that *SG affects *sentence*, not *lexical* stress, and
⁸ that this constraint does not apply in cases of given elements in narrow focus.

⁹ (5) Stress Focus (SF): Sentence stress is realized in the focus of the sentence.

¹⁰ (6) *Stress Given (*SG): Sentence stress is not realized in a given element.

¹¹ In addition to constraints regulating the relation between information structure and
¹² prosody, Czech utterances show two tendencies towards the default. These tendencies can
¹³ again be formulated as constraints (following Šimík & Wierzba 2017). The first tendency
¹⁴ is towards the canonical word order – SVO in declarative sentences – and is formulated as
¹⁵ a constraint in (7). CWO is violated when other word order variants than SVO are used.

¹⁶ (7) Canonical Word Order (CWO): The linear order of S, V, and O is SVO.

¹⁷ The second tendency is towards default (sentence) prosody. This is captured in (8) by
¹⁸ the Nuclear Stress Rule (NSR) (going back to Chomsky & Halle 1968 in the formal
¹⁹ linguistic tradition; for Czech, this was noticed already in Daneš 1957).⁴ It requires that
²⁰ the rightmost phrasal stress is assigned extra prominence – it carries the so-called nuclear
²¹ stress.

²² (8) Nuclear Stress Rule (NSR): Sentence stress is realized on the rightmost element of
²³ the sentence.

²⁴ The NSR constraint can be violated by an operation called Stress Shift (SS). Stress shift
²⁵ can be employed to stress a nonfinal focused constituent. The stress thus leaves its default
²⁶ position and moves leftwards.⁵ Importantly, the two variants (rightmost vs. shifted stress)
²⁷ do not differ in meaning and IS, only in their prosodic realization:

²⁸ (9) A: ‘Who ate my ice cream?’

³The focus-as-prominence has been challenged by the focus-as-alignment approach of Féry (2013).

⁴In this article, we make a distinction between stress and prominence. While prominence is a term connected to the prosodic realization, stress is its theoretical counterpart. Prominence is a broad term encompassing higher fundamental frequency, higher intensity, longer duration and specific intonational contours/tones.

⁵Even though we use the term “stress shift” and say that stress “moves”, it is not our intention to propose an analysis of how this happens. What is rather intended is to highlight that this operation overrides the default NSR.

- 1 B₁: Myslím, že ti zmrzlinu snědla [JARKA]_F. NSR satisfied, CWO violated
 think that you ice-cream ate Jarka
- 2 B₂: Myslím, že [JARKA]_F ti snědla zmrzlinu. NSR violated, CWO satisfied
 think that Jarka you ate ice-cream
- 3 For both: ‘I think that Jarka ate your ice cream.’

4 In the present study, the constraints Stress Focus, *Stress Given, Canonical Word Order,
 5 and Nuclear Stress Rule can be thought of as Optimality-Theoretic (OT) constraints within
 6 a weighted (rather than categorical) OT paradigm (Pater 2009; Šimík & Wierzba 2017; see
 7 also Keller 2000). This paradigm is well-suited for modeling the gradient acceptability of
 8 multiple grammatical structures with a comparable pragmatic function. Each constraint
 9 is assigned a non-binary language-specific weight, which corresponds to the “importance”
 10 of satisfying the constraint. If a language has strict word order, for instance, it has a large
 11 weight of Canonical Word Order. A violation of a constraint correlates, by assumption,
 12 with an acceptability decrease: the larger the weight, the bigger the decrease upon
 13 violation. Our hypothesis (i) assumes simple cumulativeness of violations. Our hypothesis
 14 (ii), on the other hand, includes an additional economy component, which induces an
 15 extra penalty for violating constraints in a situation where the violations could have been
 16 eschewed. If hypothesis (ii) is correct – and we will see that it is indeed supported by
 17 the data – it provides an argument for a structured approach to modeling the selection
 18 among grammatical candidates: one component is based on weighted constraints only and
 19 requires no candidate comparison; the other crucially involves candidate comparison and
 20 is activated (at least) if one of the candidates violates no constraint. An additional benefit
 21 of our study is that in addition to these competence-based effects on authenticity, we also
 22 isolate a performance-based effect: utterances involving stress shift (in violation of NSR)
 23 are more challenging to produce authentically.

24 3 Experiment

25 We designed a perception rating experiment to test the previously mentioned constraints
 26 (CWO and NSR) and their possible violations in Czech. In the following section we
 27 describe the procedure and participants, then we explain the experimental design and
 28 comment on the materials used. Finally, we present the results and discuss them.

29 3.1 Procedure and participants

30 The present experiment builds on a previous production study where we elicited and
 31 recorded utterances from 68 Czech native speakers. The stimuli were in the form of
 32 constructed dialogues, which consisted of a cue and a target. The cue utterances were

1 pre-recorded by professional actors (a female and a male), while the target utterances by
 2 the 68 native speakers. An example stimulus (dialogue) is shown in (10).

- 3 (10) A: Kdo na té oslavě s Kateřinou tančil?
 who at DEM party with Kateřina danced
 ‘Who danced with Kateřina at the party?’
 4
 5 B: Myslím, že [David]_F tančil s Kateřinou.
 think that David danced with Kateřina
 6 ‘I think that David danced with Kateřina.’

7 For the purposes of the present rating study we used a subset of the recordings from the
 8 production study, which contained the desired manipulations of CWO and NSR violations.
 9 The participants were presented with the audio stimuli, and their task was to rate the
 10 authenticity of the target utterance, i.e. the second speaker’s reaction to the first speaker.⁶
 11 To do this, participants selected a value on a Likert scale ranging from 1 (completely
 12 inauthentic) to 7 (completely authentic). Each participant rated either 94 or 82 targets in
 13 a randomized order, depending on the experimental list they were assigned. In total we
 14 had 228 participants. All of them were self-identified native Czech speakers and students
 15 of Charles University. They were compensated with ECTS credits for their participation.
 16 Before the task began, participants read instructions, agreed with the participation and
 17 tried out the rating on test stimuli. The experiment was run online on the platform
 18 PsyToolkit (Stoet, 2010, 2017).

19 We chose authenticity as the dependent variable for two reasons. Firstly, one of the
 20 goals of this study was to see how proficient our speakers from the production experiment
 21 were. We expected that their perceived authenticity as speakers would reflect their
 22 proficiency in IS encoding. The mean authenticity scores of the 68 speakers are shown in
 23 Fig. 1. Later on we will make a distinction between more proficient speakers (the top ten
 24 rated ones) and the less proficient ones (the rest), thus arriving at two speaker groups.

25 Secondly, we assume that authenticity as a variable correlates with the acceptabil-
 26 ity/naturalness of an utterance. This means that whether an utterance is perceived as more
 27 or less authentic depends not only on the speaker but also on the variables manipulated
 28 in the design. More specifically, we assumed that the authenticity of the target response
 29 was influenced by the violation of CWO, NSR, or both together.

30 3.2 Design & materials

31 The design of the main experiment was a within-participant and within-item 2×2 design.
 32 We manipulated two categorical variables: SYNTACTIC FUNCTION of the focused constituent,

⁶The participants were instructed that the cues (A) were recorded by professional actors and the responses (B) by non-professionals attempting their best possible performance.

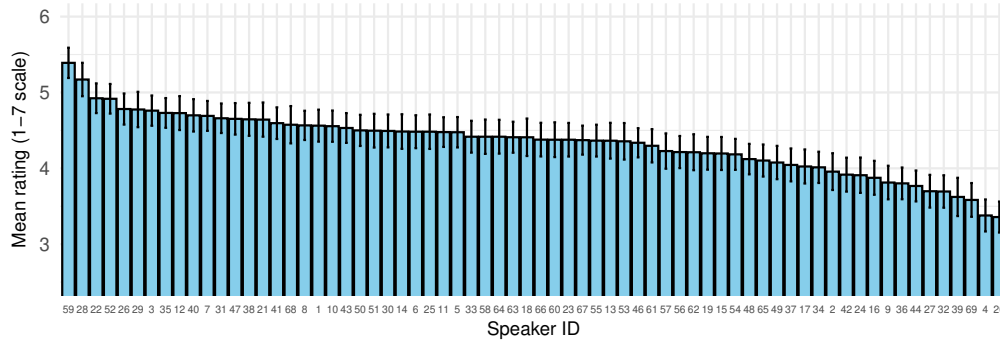


Figure 1: Mean ratings of speaker authenticity on a scale from 1 to 7 (on the y axis) for each of the 68 speakers from the production experiment (on the x axis).

1 which was either the subject or the object, and PRESENCE OF STRESS SHIFT, which was
 2 there or not. Table 1 schematically summarizes the four conditions. The expectations
 3 (Section 3.3) and statistical analysis will be formulated in terms of the satisfaction (or
 4 violation) of the two constraints under investigation – CWO and NSR, as also detailed
 5 in Table 1. CWO and NSR will thus be used as categorical (binary) predictors in the
 6 statistical models.

condition	schema	SYNT. FUNCTION	STRESS SHIFT	CWO	NSR
a	SV[O] _F	obj	no	+	+
b	[S] _F VO	subj	yes	+	–
c	OV[S] _F	subj	no	–	+
d	[O] _F VS	obj	yes	–	–

Table 1: Schematic design of the experiment; + means constraint satisfied; – constraint violated

7 Both manipulations were present in the target utterance in each dialogue, prompted
 8 by the utterance of the first speaker, as illustrated in (11). Each target contained one
 9 focused constituent (subject or object) and the background to it was always contextually
 10 given. We included focused constituents used as an answer to a *wh*-question (shown in
 11 (10)) as well as corrective foci (shown in (11)).

- 12 (11) A₁: Vypadá to, že tam servírka čepuje limonádu. for cond. a + d
 seems it that there waitress draws lemonade
 ‘It seems that that waitress there has lemonade on tap.’
 13
 14 A₂: Vypadá to, že támhleten mladík čepuje pivo. for cond. b + c
 seems it that that.DEM boy draws beer
 ‘It seems that that young guy has beer on tap.’
 15
 16 B: a. Řekl bych, že servírka čepuje [pivo]_F.
 b. Řekl bych, že [servírka]_F čepuje pivo.
 17

- 1 c. Řekl bych, že pivo čepuje [servírka]_F.
 2 d. Řekl bych, že [pivo]_F čepuje servírka.
 said AUX that beer draws waitress
 3 For all: 'I'd say that the waitress has beer on tap.'

4 Our four conditions correspond to a violation pattern concerning CWO and NSR. Condition
 5 (a) does not violate any of these two, since it has the canonical SVO word order and the
 6 stress is placed finally (i.e. it is not shifted). Condition (b) does not violate CWO, but
 7 it violates NSR by shifting the stress from the final constituent onto the subject in the
 8 initial position. Condition (c) does not violate NSR, but violates CWO. Finally, condition
 9 (d) violates both CWO and NSR as the stress shifts onto the initial object in the OVS
 10 word order.

11 The stimuli were distributed into 20 experimental lists. Each list contained utterances
 12 by 8 or 9 different target speakers, to ensure variability of the stimuli. The order of stimuli
 13 in each list was randomized. We made sure that a participant saw only one condition
 14 per item. The stimuli from the main experiment were interspersed with fillers – the
 15 filler dialogues had the same structure as the main ones, but did not involve the crucial
 16 manipulation of the independent variables described above.

17 3.3 Expectations

18 The expectations we formulate are based on the assumption that constraint violations cor-
 19 relate with perceived naturalness/acceptability and, in our specific case, with authenticity.
 20 If, for instance, the NSR is violated (if stress is shifted), we expect this to have a negative
 21 effect on authenticity rating; call it a rating drop. The magnitude of the effect correlates
 22 with the weight of the constraint violated. In a language with a small weight of CWO, for
 23 instance, noncanonical word orders will be perceived as highly acceptable.

24 Regarding the two hypotheses formulated in Section 2, they both predict rating drops
 25 upon CWO/NSR violations. Both predict the highest rating for condition (a), where no
 26 constraint is violated, and the lowest rating for condition (d), where CWO and NSR are
 27 violated at the same time. The difference between them is that hypothesis (i) predicts the
 28 drop in condition (d) to be the result of a cumulative violation of both CWO and NSR,
 29 while hypothesis (ii) predicts a more dramatic rating drop, caused by the fact that the
 30 same information structure could have been conveyed without any violation at all, as in
 31 condition (a). In terms of our statistical analysis, hypothesis (i) predicts the presence of
 32 main effects of both CWO and NSR, but no interaction between the two; hypothesis (ii)
 33 also predicts an interaction.

3.4 Results

The results were analyzed in RStudio using the package `tidyverse` Wickham et al. (2023). Statistical tests were run in RStudio using a Cumulative-Link Mixed Model (function `clmm` of the package `ordinal`; Christensen 2022) and `emmeans` (package `emmeans`; Lenth 2025). A Cumulative-Link Mixed Model is suitable for ordered categorical data (in our case this was the rating on the scale from 1 to 7). It uses the maximum likelihood estimation (MLE) which then allows for model comparison. In the model, our random effects were stimulus, participant (rater) and speaker (the voice of the target utterance), to account for their variability. The `emmeans` method counts the estimated marginal mean for each condition. The R code with the models and their results is available on OSF.⁷

The raw results of authenticity rating for the four conditions are shown in Fig. 2. The data from the total of 68 speakers were split into two groups based on their perceived proficiency: the more proficient speakers are the top 10 rated (mean rating of 4.65 or higher), the less proficient are the rest.

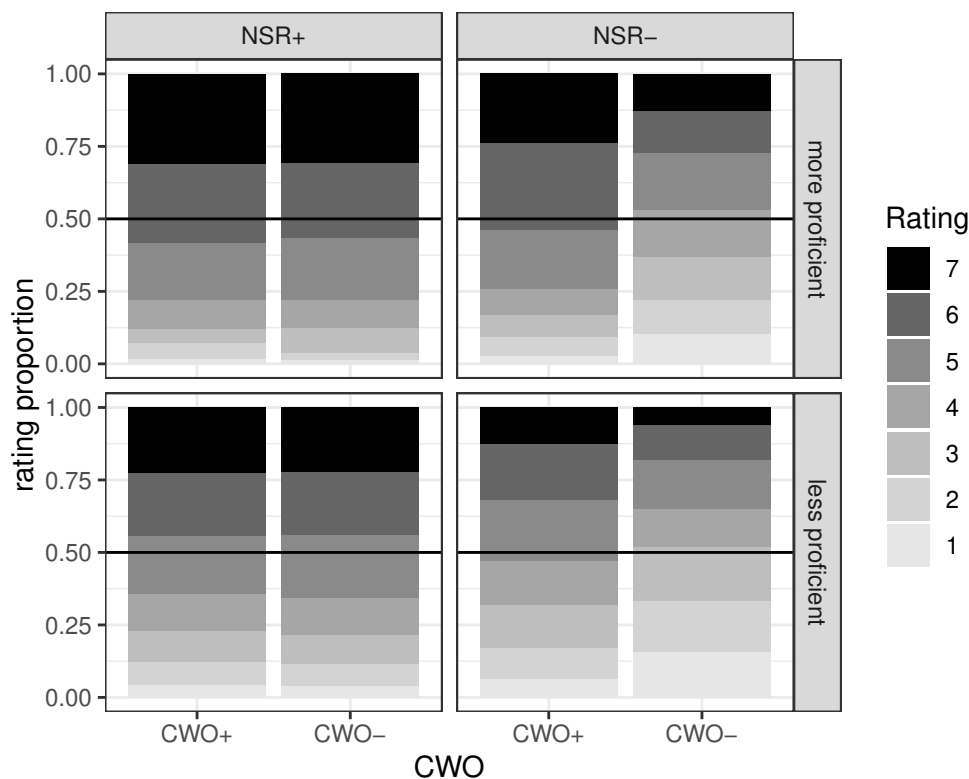


Figure 2: Ratings of the four conditions for speakers divided by their proficiency – raw results. The y-axis shows the proportion of ratings. The points of the 1-7 Likert scale are represented by the shades. The horizontal line cuts through the median for each condition.

⁷The link to the repository is here: https://osf.io/yfm8q/?view_only=759195cad0914ca28cf63dd562e9f2ac.

	Estimate	Std. Error	<i>z</i> value	<i>p</i> value
CWO	0.032 34	0.124 08	0.261	0.79
NSR	−0.589 91	0.113 92	−5.178	<0.001 ***
PROFICIENCY	0.853 71	0.241 30	3.538	<0.001 ***
CWO × NSR	−1.038 54	0.070 17	−14.800	<0.001 ***
CWO × PROFICIENCY	−0.077 30	0.193 77	−0.399	0.69
NSR × PROFICIENCY	0.011 73	0.263 70	0.044	0.97
CWO × NSR × PROFICIENCY	0.049 99	0.214 70	0.233	0.82

Table 2: Fixed effects (coefficients) of the Cumulative Link Mixed Model fitted

For this data we fitted a Cumulative-Link Mixed Model predicting the value of rating from CWO, NSR, SPEAKER PROFICIENCY and their interactions. All of these factors had two levels: CWO+ and CWO− for CWO, NSR+ and NSR− for NSR, and more proficient vs. less proficient for SPEAKER PROFICIENCY. All were treatment coded, with CWO+, NSR+, and less proficient as reference levels, respectively. We included random slopes for participants, stimuli and speakers. The resulting fixed effects (coefficients) are presented in Table 2. The interaction between CWO and NSR, visible in Table 2, was confirmed to be significant (at $p < .001$) also by comparing the model with another model excluding the relevant interaction term, using the function `anova()`.⁸ The interaction can be characterized as follows: the drop in rating caused by an NSR violation is more pronounced if CWO is violated at the same time. Next, the model reveals a main effect of PROFICIENCY ($p < .001$): responses of more proficient speakers are rated as more authentic than responses of less proficient speakers. Finally, the model shows a main effect of NSR ($p < .001$): utterances with shifted stress are rated as less authentic than utterances with default stress. A post-hoc test, based on computing the estimated marginal means (the results are visualized in Figure 3), shows that the main effect of NSR is largely contingent on the aforementioned interaction between NSR and CWO.

4 Discussion

The presence of a two-way interaction between CWO and NSR – the joint violation of CWO and NSR (condition (d)) exhibiting a rating drop that goes beyond the plain cumulation of the individual CWO/NSR violations (conditions (b) and (c)) – is consistent with the expectations drawn from hypothesis (ii). This implies that Czech speakers are not only sensitive to the markedness of noncanonical order and noncanonical stress (hypothesis (i)), but also that they implicitly contrast the uneconomical option of stress shift to a

⁸This method is called the Likelihood Ratio Test (LRT), which compares a complex model with a simpler one in order to determine whether the added complexity of the former significantly improves the fit to the data compared to the latter.

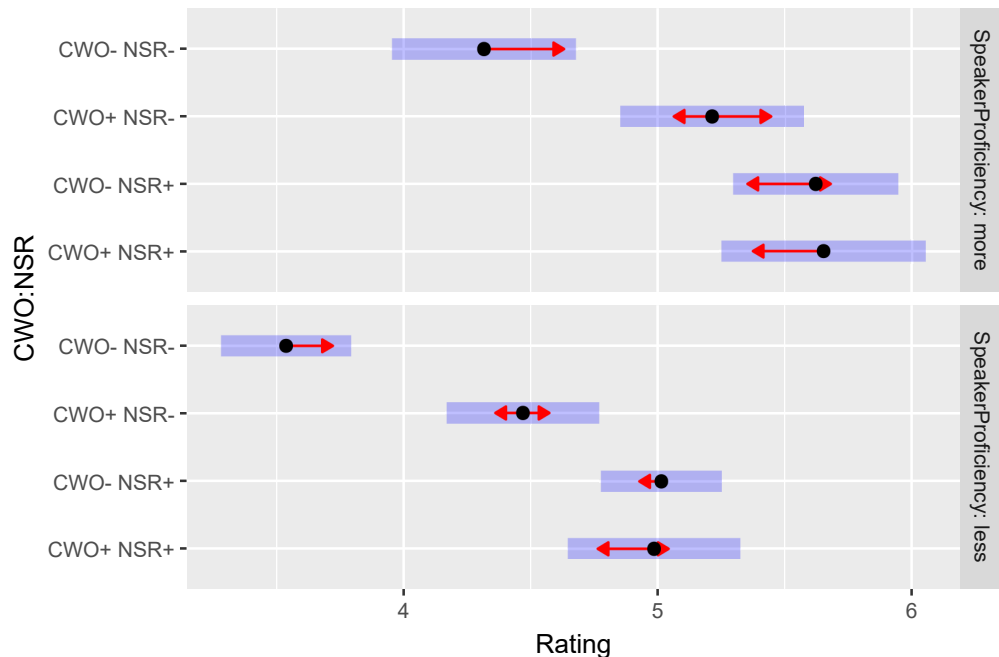


Figure 3: Ratings of the four conditions for speakers divided by their proficiency – emmeans results. The x axis shows the EMMs of ratings. The blue bars are confidence intervals for the EMMs, and the red arrows are for the comparisons among them. Whenever the red arrows overlap between two conditions, it means that there is no statistically significant difference between the two conditions.

1 fronted constituent ($[O]_F VS$) with the more economical default order and stress ($SV[O]_F$),
 2 assuming that both convey the same meaning and information structure. Furthermore, the
 3 result is similar to what Šimík & Wierzba (2017) found for non-peripheral (A-scrambled)
 4 foci. Whether the reduced acceptability (authenticity) of our $[O]_F VS$ condition and Šimík
 5 & Wierzba's $S[O]_F V$ condition (see example (3)) has a common denominator, or not,
 6 cannot be decided based on the data available so far: a single experiment comparing the
 7 two conditions would be needed. Our own intuition – expressed by the contrast between
 8 (3) and (4) – is that $[O]_F VS$ is more natural.⁹ One question that remains open is which
 9 factor would make the critical $[O]_F VS$ (or $[O]_F SV$) condition fully acceptable. Our results
 10 show that focus itself is not sufficient. We remind the reader that our items were based on
 11 two uses of focus: answerhood and corrective. However, an auxiliary analysis revealed no
 12 effect of focus use: both exhibited the same pattern.¹⁰ Among the factors to be considered

⁹The contrast might be caused by a contiguity requirement on the focus/background parts. It is possible that the structure in (3) is less acceptable, since the focused element occurs in the background domain, making it non-contiguous. In (4), the focus and background parts are contiguous, which leads to higher acceptability than (3). We thank an anonymous reviewer for this suggestion. What there is evidence for is that there seems to be no distinction between $[O]_F VS$ and $[O]_F SV$ (see Kořánová to appear). This is consistent with Šimík & Wierzba's (2017) finding that word order within the background/given/unstressed domain has no clear effect on acceptability.

¹⁰Based on these results, we expect that contrastive focus would behave in the same way as answerhood and corrective focus.

in future work are exhaustivity (aka exhaustive identification; [É. Kiss 1998](#); [Horvath 2010](#)),
 mirativity/unexpectedness ([Cruschina, 2011](#); [Trotzke, 2017](#)), and register ([Wierzba &
 Fanselow, 2020](#); [Marklová et al., 2024](#)).¹¹

Let us now move on to the individual effects of NSR and CWO. Our results reveal a
 main effect of NSR – stress shift is less authentic than default stress – but no effect of
 CWO – noncanonical order is as authentic as the canonical one. This finding, which is
 consistent with that reported in [Šimík & Wierzba \(2017\)](#), suggests that while both word
 order and prosody are flexible in Czech, the former is more flexible than the latter, or,
 in other words, that the weight of NSR is larger than that of CWO.¹² That said, the
 numerical rating difference between NSR-only violations (condition [S]_FVO) and CWO-
 only violations (condition OV[S]_F) is small compared to what we observe for NSR+CWO
 violation (condition [O]_FVS). This implies that there is no economy consideration applied
 when comparing NSR-only violations with CWO-only violations; in other words, the
 capacity of a language to change word order does not block the availability of stress shift.

Besides the expected main effect of proficiency (more proficient speakers are rated as
 more authentic than less proficient ones), post-hoc pairwise comparisons of individual
 conditions (visualized in [Figure 3](#)) within the more proficient and less proficient subsets
 reveal a notable pattern related to the authenticity of NSR-violating conditions, or, in other
 words, the naturalness of stress shift. The results of the more proficient group replicate the
 findings reported in [Groeben et al. \(2017\)](#), namely that stress shift to subjects (condition
 [S]_FVO) is as acceptable as placing subjects clause-finally (condition OV[S]_F) (see example
 (1) above).¹³ However, these two conditions differ significantly in the less proficient subset:
 the former (stress shift) condition is rated as less authentic than the latter (noncanonical
 order) condition. We take this proficiency-based difference to indicate that stress shift
 in Czech is in principle free, but it is not easy to produce in an experimental setting. If
 less proficient speaker fail to apply stress shift adequately – whether due to insufficient
 prominence on the initial constituent or due to insufficient destressing of the background
 – it leads to the violation of an information structural constraint (Stress Focus, *Stress
 Given, or both), and therefore to an additional rating drop not found in the more proficient
 group.¹⁴

¹¹In this context, it should be mentioned that already [Mathesius \(1947\)](#) noted that clause-initial focus (which he referred to as “subjective word order”) is associated with evaluative or mirative implications. Nevertheless, as far as we understand his suggestion concerned all clause-initial foci irrespective of whether they are fronted or not.

¹²In [Šimík & Wierzba \(2017\)](#), the estimated weights of NSR and CWO in their most successful “combined model” was -0.41 and -0.33 , respectively. The estimated weights correspond to the rating drop measured in standard deviations.

¹³We note that the result is replicated despite the fact that [Groeben et al. \(2017\)](#) had matrix clause stimuli, while we had embedded clause stimuli. Whether the matrix vs. embedded contrast plays any role for object focus fronting is a question for future research.

¹⁴Indeed, from the production data it is apparent that the less proficient group did not produce the

5 Conclusion

In the present article we examined the role of word order alternations and stress shift in focus encoding in Czech. Both of these operations arguably come with a cost, although stress shift seems to be more costly (Šimík & Wierzba, 2017). We compared stress shift to focused subjects vs. stress shift to fronted focused objects and expected that stress shift to a fronted object would correlate with a decrease in acceptability. The decrease was hypothesized to be a result of either a cumulative penalty on stress shift and fronting, or a penalty on using two costly operations in a situation where neither is necessary from the perspective of information structure.

To test the hypotheses, we designed a rating experiment which was based on a previous production study. In the production study, 68 Czech native speakers read utterances (targets) as reactions to pre-recorded cues. In the targets, information structure as well as word order were manipulated. Selected recordings from the production experiment were used as stimuli in the present rating experiment, where participants rated the authenticity of the target utterances. The targets in the selected stimuli featured the manipulation of the syntactic function of the focused constituent (subject vs. object) and of the presence of stress shift (present vs. not present). The resulting 4 conditions differed in the constraint violations they involved. Of particular interest was the condition [O]_FVS, which received the lowest rating. Our results suggested that this decrease in rating was caused by the categorical – rather than cumulative – effect of violating two constraints (CWO and NSR). We conclude that mere constraint cumulation is not sufficient in accounting for the acceptability of focused object fronting in Czech; instead, the reduced acceptability is a result of a global comparison between candidates: the costly [O]_FVS structure is penalized because the free SV[O]_F could have been used.

Additionally, our data showed that shifting stress in Czech is possible (in terms of the system/competence), but not necessarily easy (in terms of performance). The latter point was demonstrated on the ratings of the more proficient vs. less proficient speakers, where the less proficient group had difficulties in producing the expected stress shift. This underscores the conclusion that while both word order and prosody are flexible in Czech (Vallduví & Engdahl, 2013; Šimík & Wierzba, 2017), the former is more flexible than the latter.

prosodic difference (in terms of F0) between the initial focus and the following background to such an extent as the more proficient group. It is not in the scope of this article to present the full analysis of this issue, though.

References

- Alter, Kai. 1997a. *Formale Slavistik* chap. Fokusprosodie im Russischen: Phonologische und akustische Korrelate von Informationsstrukturierung, 399–414. Frankfurt am Main: Vervuert.
- Alter, Kai. 1997b. Russian prosody: Phrasing and tonal structure. In M Lindseth & S Franks (eds.), *Formal Approaches to Slavic Linguistics 5. The Indiana Meeting, 1996*, 1–17. Ann Arbor, MI: Michigan Slavic Publications, .
- Baumann, Stefan, Martine Grice & Susanne Steindamm. 2006. Prosodic marking of focus domains - categorical or gradient? In *Speech Prosody 2006*, paper 065. ISCA. doi:10.21437/speechprosody.2006-73.
- Baumann, Stefan & Arndt Riestler. 2013. Coreference, lexical givenness and prosody in German. *Lingua* 136. 16–37. doi:10.1016/j.lingua.2013.07.012.
- Breen, Mara, Evelina Fedorenko, Michael Wagner & Edward Gibson. 2010. Acoustic correlates of information structure. *Language and Cognitive Processes* 25(7–9). 1044–1098. doi:10.1080/01690965.2010.504378.
- Chomsky, Noam & Morris Halle. 1968. *The Sound Pattern of English*. New York, NY: Harper and Row.
- Christensen, Rune Haubo Bojesen. 2022. ordinal: Regression models for ordinal data. <https://CRAN.R-project.org/package=ordinal>.
- Cruschina, Silvio. 2011. *Discourse-related features and functional projections*. Oxford: Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199759613.001.0001>.
- Daneš, František. 1957. *Intonace a věta ve spisovné češtině*. Prague: Československá akademie věd.
- É. Kiss, Katalin. 1998. Identificational focus versus information focus. *Language* 74(2). 245–273. <https://doi.org/10.1353/lan.1998.0211>.
- Firbas, Jan. 1992. *Functional sentence perspective in written and spoken communication*. Cambridge University Press. doi:10.1017/cbo9780511597817.
- Féry, Caroline. 2013. Focus as prosodic alignment. *Natural Language & Linguistic Theory* 31(3). 683–734. doi:10.1007/s11049-013-9195-7.

- 1 Groeben, Lena, Radek Šimík & Frank Kügler. 2017. Stress shift, focus, and givenness in
2 Czech. In Yohei Oseki, Masha Esipova & Stephanie Harves (eds.), *Formal Approaches*
3 *to Slavic Linguistics (FASL) 24: The New York University Meeting 2015*, 180–199. Ann
4 Arbor, MI: Michigan Slavic Publications.
- 5 Hamlaoui, Fatima, Marzena Żygis, Jonas Engelmann & Michael Wagner. 2018. Acoustic
6 correlates of focus marking in Czech and Polish. *Language and Speech* 62(2). 358–377.
7 doi:10.1177/0023830918773536.
- 8 Horvath, Julia. 2010. “Discourse features”, syntactic displacement, and the status of
9 contrast. *Lingua* 120(6). 1346–1369. doi:10.1016/j.lingua.2008.07.011.
- 10 Jackendoff, Ray. 1972. *Semantic interpretation in generative grammar*. Cambridge: MIT
11 Press.
- 12 Keller, Frank. 2000. *Gradience in grammar*. Edinburgh: University of Edinburgh disserta-
13 tion. <https://hdl.handle.net/1842/744>.
- 14 Kořánová, Daniela. to appear. *Prosodic encoding of contrastive topic in Czech*. Charles
15 University MA thesis.
- 16 Krifka, Manfred. 2008. Basic notions of information structure. *Acta Linguistica Hungarica*
17 55(3-4). 243–276. <https://www.jstor.org/stable/26190905>.
- 18 Kučerová, Ivona. 2011. Grammatical marking of givenness. *Natural Language Semantics*
19 20(1). 1–30. doi:10.1007/s11050-011-9073-y.
- 20 Kügler, Frank & Anja Gollrad. 2015. Production and perception of contrast: The case of
21 the rise-fall contour in German. *Frontiers in Psychology* 6. doi:10.3389/fpsyg.2015.01254.
- 22 Lenerz, Jürgen. 1977. *Zur Abfolge nominaler Satzglieder im Deutschen* (Studien zur
23 deutschen Grammatik 5). Tübingen: Gunter Narr.
- 24 Lenth, Russell V. 2025. *emmeans: Estimated Marginal Means, aka Least-Squares Means*.
25 <https://rvlenth.github.io/emmeans/>. R package version 1.11.2-8.
- 26 Marklová, Anna, Alžběta Houzar, Roland Meyer, Olga Buchmüller, Chris Demian & Luka
27 Szucsich. 2024. Left dislocations and long topicalizations in Czech: An acceptability
28 judgements study. *Zeitschrift für Slawistik* 69(3). 553–574. [https://doi.org/10.1515/](https://doi.org/10.1515/slawn-2024-0026)
29 [slawn-2024-0026](https://doi.org/10.1515/slawn-2024-0026).
- 30 Mathesius, Vilém. 1939. O tak zvaném aktuálním členění větném. *Slovo a slovesnost* 5(4).
31 171–174.

- 1 Mathesius, Vilém. 1947. *Čeština a obecný jazykozpyt*. Praha: Melantrich.
- 2 Pater, Joe. 2009. Weighted constraints in generative linguistics. *Cognitive Science* 33(6).
3 999–1035. doi:10.1111/j.1551-6709.2009.01047.x.
- 4 Reinhart, Tanya. 2006. *Interface strategies: Optimal and costly computations*. The MIT
5 Press. doi:10.7551/mitpress/3846.001.0001.
- 6 Rooth, Mats. 1992. A theory of focus interpretation. *Natural Language Semantics* 1(1).
7 75–116. doi:10.1007/bf02342617.
- 8 Sgall, P., E. Hajičová & E. Buráňová. 1980. *Aktuální členění věty v češtině*. Prague:
9 Academia.
- 10 Siewierska, Anna & Ludmila Uhlířová. 1998. *An overview of word order in Slavic languages*
11 105–150. De Gruyter Mouton. doi:10.1515/9783110812206.105.
- 12 Šimík, Radek. 2024. Prosodic reflexes of information structure. In Danko Šipka & Wayles
13 Browne (eds.), *The Cambridge handbook of Slavic linguistics* Cambridge Handbooks
14 in Language and Linguistics, 104–126. Cambridge: Cambridge University Press. doi:
15 10.1017/9781108973021.007.
- 16 Šimík, Radek & Marta Wierzba. 2017. Expression of information structure in West Slavic:
17 Modeling the impact of prosodic and word-order factors. *Language* 93(3). 671–709.
18 <https://muse.jhu.edu/article/669550>.
- 19 Stoet, Gijsbert. 2010. PsyToolkit: a software package for programming psychological
20 experiments using Linux. *Behavior Research Methods* 42(4). 1096–1104. doi:10.3758/
21 BRM.42.4.1096.
- 22 Stoet, Gijsbert. 2017. PsyToolkit: A novel web-based method for running online ques-
23 tionnaires and reaction-time experiments. *Teaching of Psychology* 44(1). 24–31. doi:
24 10.1177/0098628316677643. Place: US.
- 25 Trotzke, Andreas. 2017. Mirative fronting in German: Experimental evidence. *Review of*
26 *Cognitive Linguistics* 15(2). 460–488. <https://doi.org/10.1075/rcl.15.2.07tro>.
- 27 Truckenbrodt, Hubert. 1995. *Phonological phrases: Their relation to syntax, focus and*
28 *prominence*. Cambridge: MIT dissertation.
- 29 Vallduví, Enric & Elisabet Engdahl. 2013. The linguistic realization of information
30 packaging. *Linguistics* 51(1). 19–20. <https://doi.org/10.1515/ling-2013-0041>.

- 1 Wackernagel, J. 1892. Über ein Gesetz der indogermanischen Wortstellung. *Indogermanis-*
2 *che Forschungen* 1(1). 333–436. <https://doi.org/10.1515/9783110242430.333>.
- 3 Wickham, Hadley, Davis Vaughan & Maximilian Girlich. 2023. tidy: Tidy messy data.
4 <https://github.com/tidyverse/tidy>.
- 5 Wierzba, Marta & Gisbert Fanselow. 2020. Factors influencing the acceptability of object
6 fronting in German. *The Journal of Comparative Germanic Linguistics* 23(1). 77–124.
7 <https://doi.org/10.1007/s10828-020-09113-1>.
- 8 Zubizarreta, Maria Luisa. 1998. *Prosody, focus and word order*. Cambridge: MIT Press.
- 9 Zybatow, Gerhild & Grit Mehlhorn. 2000. Experimental evidence for focus structure in
10 Russian. In Tracy Holloway King & Irina A. Sekerina (eds.), *Formal Approaches to*
11 *Slavic Linguistics 8: The Philadelphia Meeting 1999*, 414–434. Ann Arbor, MI: Michigan
12 Slavic Publications.